

Software Engineering

Software Project Management

It's a practice of engineering, not just coding.

Chapter: 24, 28

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Introduction to Software Project Management

- Software Project Management (SPM) is the process of planning, organizing, monitoring, and controlling software projects.
- It helps software teams complete projects successfully within time, budget, and quality requirements.
- SPM combines technical knowledge, communication skills, teamwork, and management practices.
- It is widely used in software companies, startups, research projects, and academic development teams.

Importance of Software Project Management

- Ensures software projects are completed on schedule.
- Helps manage project cost and resources effectively.
- Improves software quality and customer satisfaction.
- Reduces development risks and unexpected failures.
- Supports communication among developers, testers, clients, and managers.

Software Development Life Cycle (SDLC)

- Requirement Analysis: Gather and analyze customer requirements.
- Planning: Estimate time, cost, and resources.
- Design: Create software architecture and system design.
- Development: Write and implement program code.
- Testing: Detect and fix software bugs.
- Deployment: Release software to users.
- Maintenance: Update and improve the software after release.

Software Development Methodologies

- Waterfall Model: Sequential development process.
- Agile Methodology: Flexible and iterative development.
- Scrum: Agile framework using sprints and daily meetings.
- DevOps: Combines development and operations for continuous delivery.
- Spiral Model: Focuses on risk analysis and iterative development.

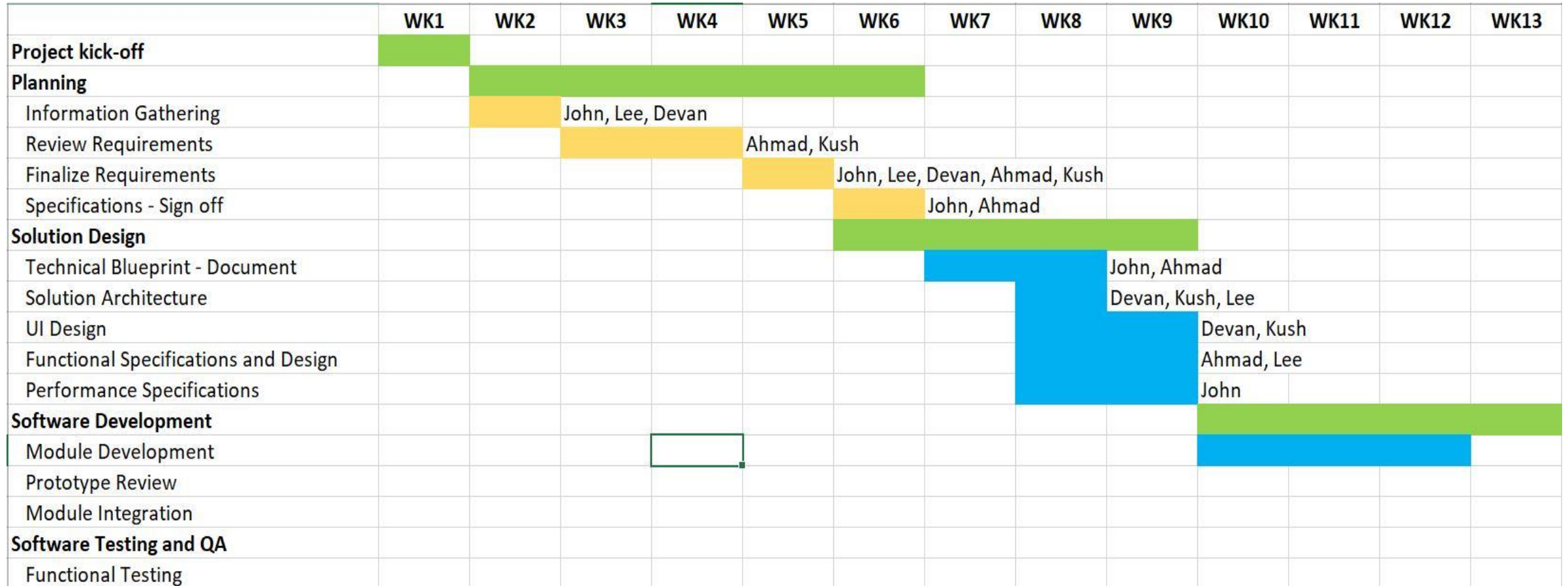
Project Planning

- Project planning defines the roadmap for software development.
- Managers identify project scope, objectives, timeline, and deliverables.
- Resources such as developers, tools, and budget are allocated.
- Effective planning reduces confusion and project delays.

Project Scheduling

- Scheduling determines when tasks will be completed.
- Important scheduling techniques include:
 - ***Gantt Chart*** – Visual timeline of project activities.
 - ***PERT Chart*** – Shows task dependencies and estimated durations.
 - ***Critical Path Method (CPM)*** – Identifies the longest sequence of tasks.
 - ***Agile Sprint Planning*** – Divides work into short iterations.

Gantt Chart



Cost Estimation

- Cost estimation predicts the budget required for the project.
- Includes developer salary, hardware, software, testing, and maintenance costs.
- *Common estimation techniques:*
 - Expert Judgment
 - COCOMO Model
 - Function Point Analysis

Risk Management

- Risk management identifies and handles possible project problems.
- Common software risks include:
 - Requirement changes
 - Team member unavailability
 - Technical failures
 - Security vulnerabilities
- Risk mitigation strategies help reduce project failure.

Team Management

- Effective teamwork is essential for project success.
- Project managers assign roles and responsibilities.
- Good communication improves collaboration and reduces conflict.
- Motivation and leadership increase team productivity.
- Performance monitoring helps track project progress.

Version Control System

- Version Control Systems (VCS) manage changes in source code.
- They help multiple developers work on the same project safely.
- VCS tracks file history, modifications, and collaboration.
- Popular version control tools include ***Git and GitHub.***

Git & GitHub

- Git is a distributed version control system.
- Developers can track code changes and restore previous versions.
- GitHub is a cloud platform for hosting Git repositories.
- It supports collaboration among developers worldwide.
- Features of GitHub:
 - Repository hosting
 - Issue tracking
 - Pull requests
 - Project boards
 - GitHub Actions for CI/CD

Branching and Merging in Git

- Branches allow developers to work independently on features.
- Main branch usually contains stable code.
- Feature branches are created for new functionality.
- Merging combines code changes into the main branch.
- Git helps avoid conflicts during collaborative development.

Project Management Tools

- Jira – Agile project and issue management.
- ***Trello – Kanban-based task management.***
- ClickUp – All-in-one collaboration platform.
- GitHub Projects – Development tracking integrated with GitHub.
- Docker – Container-based development and deployment.

Docker in Software Projects

- Docker packages applications with dependencies into containers.
- Ensures software runs consistently on different machines.
- Helps simplify deployment and testing.
- Supports microservices architecture and cloud deployment.
- Useful for web, AI, and research-based software projects.

Software Quality Assurance

- Quality assurance ensures the software meets requirements.
- Testing types include:
 - Unit Testing
 - Integration Testing
 - System Testing
 - Acceptance Testing
- Automation tools improve testing efficiency.

Challenges in Software Project Management

- Changing customer requirements
- Time and budget limitations
- Communication gaps among stakeholders
- Technical complexity
- Resource shortages
- Managing distributed teams

Best Practices

- Use Agile methodologies for flexibility.
- Maintain proper documentation.
- Use Git and GitHub for collaboration.
- Regularly monitor project progress.
- Conduct ***code reviews and testing***.
- Maintain effective communication.

Conclusion

- Software Project Management is essential for successful software development.
- It improves planning, collaboration, quality, and delivery.
- Modern tools like Git, GitHub, Docker, and Jira/Trello simplify project management.
- Effective management increases software reliability and customer satisfaction.

Thank you